

(2)

Ex 3.2

Q1 Show that inverse of diagonal Matrix with all non-zero diagonal elements is a diagonal matrix.

Sol Let $A = \begin{pmatrix} d_1 & 0 & 0 & \dots & 0 \\ 0 & d_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_n \end{pmatrix}$ be a diagonal Matrix of order 'n'

and Let $B = \begin{pmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & b_{22} & b_{23} & \dots & b_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & b_{n3} & \dots & b_{nn} \end{pmatrix}$ be the inverse of diagonal Matrix 'A'.

By def of Inverse $AB = I_n$

$$\therefore \begin{pmatrix} d_1 & 0 & 0 & \dots & 0 \\ 0 & d_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & d_n \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & b_{22} & b_{23} & \dots & b_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & b_{n3} & \dots & b_{nn} \end{pmatrix} = I_n$$

$$\begin{pmatrix} d_1 b_{11} & d_1 b_{12} & \dots & d_1 b_{1n} \\ d_2 b_{21} & d_2 b_{22} & \dots & d_2 b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_n b_{n1} & d_n b_{n2} & \dots & d_n b_{nn} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

$$\Rightarrow d_i b_{ij} = 0 \quad \Rightarrow b_{ij} = \frac{0}{d_i} = 0$$

for $i \neq j = 1, 2, \dots, n$

$$\Rightarrow d_i b_{ii} = 1 \quad \Rightarrow b_{ii} = \frac{1}{d_i}$$

for $i = 1, 2, \dots, n$

Hence.

$$B = \begin{pmatrix} \frac{1}{d_1} & 0 & 0 & \dots & 0 \\ 0 & \frac{1}{d_2} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \frac{1}{d_n} \end{pmatrix}$$

is the diagonal Matrix
Inverse of diagonal Matrix A is diagonal Matrix B.

Q2 Show that inverse of a scalar Matrix is a scalar Matrix.

Sol

Let $A = \begin{pmatrix} c & 0 & 0 & \dots & 0 \\ 0 & c & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & c \end{pmatrix}$ be a Scalar Matrix of order 'n'

and

$B = \begin{pmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & b_{22} & b_{23} & \dots & b_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & b_{n3} & \dots & b_{nn} \end{pmatrix}$ be the inverse of scalar Matrix A

By def of Inverse

$$AB = I_n$$

$$\therefore \begin{pmatrix} c & 0 & 0 & \dots & 0 \\ 0 & c & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & c \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & b_{22} & b_{23} & \dots & b_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & b_{n3} & \dots & b_{nn} \end{pmatrix} = I_n$$

$$\begin{pmatrix} b_{11}c & b_{12}c & \dots & b_{1n}c \\ b_{21}c & b_{22}c & \dots & b_{2n}c \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1}c & b_{n2}c & \dots & b_{nn}c \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

$$\Rightarrow b_{ij}c = 0 \quad \Rightarrow b_{ij} = \frac{0}{c} = 0$$

for $i \neq j = 1, 2, \dots, n$

$$\Rightarrow b_{ii}c = 1 \quad \Rightarrow b_{ii} = \frac{1}{c}$$

for $i = 1, 2, \dots, n$

Hence

$$B = \begin{pmatrix} \frac{1}{c} & 0 & 0 & \dots & 0 \\ 0 & \frac{1}{c} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \frac{1}{c} \end{pmatrix}$$

is the Scalar Matrix
 $\therefore$ Inverse of Scalar Matrix is Scalar.

② (i) $(A^n)^{-1} = (\bar{A}^{-1})^n$ To Prove
 Sol: $(A^n)^{-1} = (A \cdot A \cdot A \dots n \text{ times})^{-1}$
 $= \bar{A}^{-1} \bar{A}^{-1} \bar{A}^{-1} \dots n \text{ times}$
 $(A^n)^{-1} = (\bar{A}^{-1})^n$ proved.

(iv) $(\bar{A})^{-1} = \overline{(A^{-1})}$
 $(\bar{A})(\bar{A}^{-1}) = \overline{(A A^{-1})} = \bar{I} = I$
 Pre Multiply by $(\bar{A})^{-1}$
 $(\bar{A})^{-1}(\bar{A})(\bar{A}^{-1}) = (\bar{A})^{-1}I$
 $I(\bar{A}^{-1}) = (\bar{A})^{-1}$
 $(\bar{A}^{-1}) = \overline{(A^{-1})}$

(v) $(\bar{A}^t)^{-1} = \overline{(A^{-1})^t}$
 $(\bar{A}^t)(\bar{A}^{-1})^t = \overline{A^t(A^{-1})^t} = \overline{(A^t A^{-1})^t} = \overline{(I)^t} = \bar{I} = I$
 Pre multiply both sides by $(\bar{A}^t)^{-1}$
 $(\bar{A}^t)^{-1}(\bar{A}^t)(\bar{A}^{-1})^t = (\bar{A}^t)^{-1}I$
 $(\bar{A}^{-1})^t = \overline{(A^t)^{-1}}$ proved.

(iii) $(A^{-1})^t = (A^t)^{-1}$ To Prove
 Sol: $(\bar{A}^{-1})^t(A^t) = \overline{(A A^{-1})^t} = \bar{I}^t = I$
 $(\bar{A}^{-1})^t(I) = (A^t)^{-1}$ Post x by $(A^t)^{-1}$
 $(\bar{A}^{-1})^t = (A^t)^{-1}$ proved.

(vi) $(KA)^{-1} = K^{-1}A^{-1}$ To Prove
 Sol: $(KA)(K^{-1}A^{-1}) = (AK)(K^{-1}A^{-1})$
 $= A(KK^{-1})A^{-1}$
 $= AIA^{-1}$
 $= AA^{-1}$
 $= I$
 $(KA)^{-1}(KA)(K^{-1}A^{-1}) = (KA)^{-1}I$ Pre Multiply by $(KA)^{-1}$
 $(I)K^{-1}A^{-1} = (KA)^{-1}$
 $K^{-1}A^{-1} = (KA)^{-1}$

Q4 Given that $A^2B = B^2A$ — (i) and $A^3 = B^3$ — (ii), $A \neq B$ and $A, B \neq 0$

Note

Suppose A^2+B^2 is invertible
 then $A-B = I(A-B)$ $\because I$ is identity Matrix
 $= (A^2+B^2)^{-1}(A^2+B^2)(A-B)$ $\because I = (A^2+B^2)^{-1}(A^2+B^2)$
 $= (A^2+B^2)^{-1}(A^3-A^2B+B^2A-B^3)$
 $= (A^2+B^2)^{-1}(0)$ using (i) & (ii)

$A-B = 0$
 $\Rightarrow A = B$, a contradiction as $A \neq B$ are given diff matrices.
 Hence our supposition is wrong and A^2+B^2 is not invertible.

$(A^t)^{-1} = (\bar{A}^{-1})^t$
 $(\bar{A}^{-1}) = \overline{(A^t)^{-1}}$
 $(\bar{A}^{-1})^n = (\bar{A}^{-1})^n$
 $(\bar{A}^t) = \overline{(A^t)^t}$
 $(\bar{A}^t)^n = (\bar{A}^t)^n$
 $(\bar{A}^n) = \overline{(A^n)^t}$
 All these inverse, transpose conjugate, exponent are interchangeable.

Q3. If A is invertible & $AB = 0$ show that $B = 0$

Ans. A is invertible so A^{-1} exists

$$AB = 0 \text{ given}$$

$$A^{-1}AB = A^{-1}0$$

$$IB = 0$$

$$B = 0$$

Elementary Row Operations (ERO)

- (i) These operations on a matrix are called ERO
- (i) Interchange of any two rows denoted by R_{ij} i.e. i th & j th rows are interchanged.
- (ii) Multiplication of a row by a scalar denoted by $K(R_i)$ i.e. each element of i th row is multiplied by K .
- (iii) Addition of a multiple of one row to any other row, denoted by $(K)R_i + R_j$.

Row Equivalent Matrices

A matrix B is said to be row equivalent to a matrix A of same order if B can be obtained from A by applying finite no of ERO on A .

$B \sim A$ B is row equivalent to A .

Elementary Matrix

The matrix E obtained by applying one ERO to I is called an Elementary Matrix.

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & K \end{bmatrix} \text{ by } KR_3$$

Echelon Form of a Matrix

A matrix is said to be in Echelon form if it has the following structure.

- i) All the zero rows are below the non-zero rows of A.
- ii) The number of zeros occurring before the first non-zero entry in each non-zero row is greater than the number of zeros that appear before the first non-zero element in any preceding row.

For example

$$\begin{bmatrix} 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 5 \end{bmatrix}, \begin{bmatrix} 0 & -1 & -2 \end{bmatrix} \quad A = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 1 & 2 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 4 & 9 \\ 0 & 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad H = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

A, B, C are not in Row Echelon Form.
D, E, F, G, H are in Reduced Echelon Form.
Rest are Echelon Form.

$$D = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \quad E = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad F = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad G = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

The first non-zero element in each row of an Echelon Matrix Form is called Pivot element of that row. A column containing Pivot is called Pivot Column.

Reduced Echelon Form has the following structure.

- i) It is in Echelon Form
- ii) Pivot element of each row is '1' i.e. first non-zero element is '1' in each row.
- iii) Every entry in the pivot column is zero except the pivot element '1'.

Note It becomes easy to reduce a matrix in Echelon form if we obtain first element to be '1'. Also some authors include this condition in Echelon form instead of Reduced Echelon form.
Echelon form is different for a matrix depending upon the sequence of ERO applied but Reduced Echelon form is the same.

If a square matrix A is reduced to the identity matrix by a sequence of elementary ^{row} operations, the same sequence of operations performed on the identity matrix produces the inverse of A .

Proof

$$I \sim A$$

$$\Rightarrow I = (E_n E_{n-1} E_{n-2} \dots E_2 E_1) A$$

where E_i are suitable elementary matrices.

Post Multiply both sides by A^{-1}

$$I A^{-1} = (E_n E_{n-1} E_{n-2} \dots E_2 E_1) A A^{-1}$$

$$A^{-1} = (E_n E_{n-1} E_{n-2} \dots E_2 E_1) I$$

x ----- x

Q4 Find the inverse of

(i) $\begin{bmatrix} 1 & 0 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ over R



Matrix A

$$\begin{bmatrix} 1 & 0 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

I_3

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reducing A to Identity matrix by ERO.

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

I_3

$$\begin{bmatrix} 1 & 0 & -k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

by $-kR_3 + R_1$

Hence $\begin{bmatrix} 1 & 0 & -k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is the inverse of A

$$(ii) A = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \text{ by } R_{13}$$

$$R \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{ by } R_{23}$$

$$R \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{ by } (-1)R_2$$

$$R \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \text{ by } (-1)R_3$$

 I_3

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Hence $\begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$ is inverse of matrix A.

x ————— x

$$(iii) A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R \begin{bmatrix} 1 & -2 & 3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ by } (-1)R_1$$

$$R \begin{bmatrix} 1 & -2 & 3 \\ 0 & 5 & -6 \\ 0 & 6 & -7 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix} \text{ by } \begin{matrix} -2R_1 + R_2 \\ -4R_1 + R_3 \end{matrix}$$

$$\sim \begin{pmatrix} 1 & -2 & 3 \\ 0 & 5 & -6 \\ 0 & 1 & -1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix}$$

 $-R_2 + R_3$

$$\sim \begin{pmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \\ 0 & 5 & -6 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 0 \\ 2 & -1 & 1 \\ 2 & 1 & 0 \end{pmatrix}$$

 R_{23}

$$\sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 3 & -2 & 2 \\ 2 & -1 & 1 \\ -8 & 6 & -5 \end{pmatrix}$$

 $2R_2 + R_1$
 $-5R_2 + R_3$

$$\sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 3 & -2 & 2 \\ 2 & -1 & 1 \\ 8 & -6 & 5 \end{pmatrix}$$

 $(-1)R_3$

$$\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -5 & 4 & -3 \\ 10 & -7 & 6 \\ 8 & -6 & 5 \end{pmatrix}$$

 $-R_3 + R_1$
 $R_3 + R_2$
 I_3

Hence $\begin{pmatrix} -5 & 4 & -3 \\ 10 & -7 & 6 \\ 8 & -6 & 5 \end{pmatrix} = \bar{A}^{-1}$

(iv) $A = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{pmatrix}$

$$\frac{I}{3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\sim \begin{pmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 1 & 0 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

 $-2R_1 + R_3$

$$\sim \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 2 & 1 & -1 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

 R_{13}

$$R \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 0 & 1 & +1 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 0 & 1 \\ 0 & 1 & 0 \\ 5 & 0 & -2 \end{pmatrix}$$

 $-2R_1 + R_3$

$$R \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 0 & 1 \\ 5 & 0 & -2 \\ 0 & 1 & 0 \end{pmatrix}$$

 R_{23}

$$R \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 0 & 1 \\ 5 & 0 & -2 \\ -10 & 1 & 4 \end{pmatrix}$$

 $-2R_2 + R_3$

$$R \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 0 & 1 \\ 5 & 0 & -2 \\ 10 & -1 & -4 \end{pmatrix}$$

 $(-1)R_3$

$$R \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ -1 & -4 & \end{pmatrix}$$

 $R_3 + R_1$
 $-R_3 + R_2$
 I_3

Hence $\begin{pmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{pmatrix} = A^{-1}$

(vi) $A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$$I_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$R \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} R_{13}$$

 I_4

Hence $\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = A^{-1}$

$$A = \begin{pmatrix} i & -1 & 2i \\ 2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R \begin{pmatrix} 1 & i & 2 \\ 2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -i & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

 $(-i)R_1$

$$R \begin{pmatrix} 1 & i & 2 \\ 0 & -2i & -2 \\ 0 & i & 3 \end{pmatrix}$$

$$\begin{pmatrix} -i & 0 & 0 \\ 2i & 1 & 0 \\ -i & 0 & 1 \end{pmatrix}$$

 $-2R_1 + R_2$
 $R_1 + R_3$

$$R \begin{pmatrix} 1 & i & 2 \\ 0 & 1 & -i \\ 0 & i & 3 \end{pmatrix}$$

$$\begin{pmatrix} -i & 0 & 0 \\ -1 & i/2 & 0 \\ -i & 0 & 1 \end{pmatrix}$$

 $(\frac{i}{2})R_2$

$$R \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -i \\ 0 & 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & \frac{1}{2} & 0 \\ -1 & i/2 & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

 $-iR_2 + R_1$
 $-iR_2 + R_3$

$$R \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -i \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & \frac{1}{2} & 0 \\ -1 & i/2 & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$$

 $\frac{1}{2} R_3$

$$R \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & \frac{1}{4} & -\frac{1}{2} \\ -1 & \frac{3i}{4} & \frac{i}{2} \\ 0 & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$$

 $-R_3 + R_1$
 $iR_3 + R_2$
 I_3

Hence $A^{-1} = \begin{pmatrix} 0 & \frac{1}{4} & -\frac{1}{2} \\ -1 & \frac{3i}{4} & \frac{i}{2} \\ 0 & \frac{1}{4} & \frac{1}{2} \end{pmatrix}$

$$⑥ \quad (i) \begin{pmatrix} 1 & -2 & 3 & -1 \\ 2 & -1 & 2 & 2 \\ 3 & 1 & 2 & 3 \end{pmatrix}$$

$$R \begin{pmatrix} 1 & -2 & 3 & -1 \\ 0 & 3 & -4 & 4 \\ 0 & 7 & -7 & 6 \end{pmatrix} \quad \begin{array}{l} -2R_1 + R_2 \\ -3R_1 + R_3 \end{array}$$

$$R \begin{pmatrix} 1 & -2 & 3 & -1 \\ 0 & 1 & -\frac{4}{3} & \frac{4}{3} \\ 0 & 7 & -7 & 6 \end{pmatrix} \quad \frac{1}{3} R_2$$

$$R \begin{pmatrix} 1 & 0 & \frac{1}{3} & \frac{5}{3} \\ 0 & 1 & -\frac{4}{3} & \frac{4}{3} \\ 0 & 0 & 7/3 & -1\frac{1}{3} \end{pmatrix} \quad \begin{array}{l} 2R_2 + R_1 \\ -7R_2 + R_3 \end{array}$$

$$R \begin{pmatrix} 1 & 0 & \frac{1}{3} & \frac{5}{3} \\ 0 & 1 & -\frac{4}{3} & \frac{4}{3} \\ 0 & 0 & 1 & -1\frac{1}{7} \end{pmatrix} \quad \frac{3}{7} R_3$$

$$R \begin{pmatrix} 1 & 0 & 0 & 15/7 \\ 0 & 1 & 0 & -4/7 \\ 0 & 0 & 1 & -10/7 \end{pmatrix} \quad \begin{array}{l} \frac{4}{3} R_3 + R_2 \\ -\frac{1}{3} R_3 + R_1 \end{array}$$

Required Reduced Echelon Form

$$A = \begin{pmatrix} 0 & 1 & 3 & -2 \\ 2 & 1 & -4 & 3 \\ 2 & 3 & 2 & -1 \end{pmatrix}$$

$$B \begin{pmatrix} 2 & 1 & -4 & 3 \\ 0 & 1 & 3 & -2 \\ 2 & 3 & 2 & -1 \end{pmatrix} \quad R_{12}$$

$$\sim R \begin{pmatrix} 1 & \frac{1}{2} & -2 & \frac{3}{2} \\ 0 & 1 & 3 & -2 \\ 2 & 3 & 2 & -1 \end{pmatrix} \quad \frac{1}{2} R_1$$

$$\sim R \begin{pmatrix} 1 & \frac{1}{2} & -2 & \frac{3}{2} \\ 0 & 1 & 3 & -2 \\ 0 & 2 & 6 & -4 \end{pmatrix} \quad -2R_1 + R_3$$

$$\sim R \begin{pmatrix} 1 & 0 & -7/2 & 5/2 \\ 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} -\frac{1}{2}R_2 + R_1 \\ -2R_2 + R_3 \end{matrix}$$



Required Reduced Echelon Form

$$(iii) A = \begin{pmatrix} 1 & 2 & -1 & 2 & 1 \\ 2 & 4 & 1 & -2 & 3 \\ 3 & 6 & 2 & -6 & 3 \end{pmatrix}$$

$$\sim R \begin{pmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 3 & -6 & 1 \\ 0 & 0 & 5 & -12 & 0 \end{pmatrix} \quad \begin{matrix} -2R_1 + R_2 \\ -3R_3 + R_2 \end{matrix}$$

$$\sim R \begin{pmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 1 & -2 & \frac{1}{3} \\ 0 & 0 & 5 & -12 & 0 \end{pmatrix} \quad \frac{1}{3} R_2$$

$$\sim R \begin{pmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 1 & -2 & \frac{1}{3} \\ 0 & 0 & 0 & -2 & -\frac{5}{3} \end{pmatrix} \quad -5R_2 + R_3$$

$$\sim R \begin{pmatrix} 1 & 2 & -1 & 2 & 1 \\ 0 & 0 & 1 & -2 & \frac{1}{3} \\ 0 & 0 & 0 & -2 & \frac{5}{6} \end{pmatrix} \quad -\frac{1}{2} R_3$$

Echelon Form.

$$(iv) A = \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 2 & -5 & 3 & 1 \\ 4 & 1 & 1 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 11 & -5 & 3 \\ 0 & -11 & 5 & -3 \\ 0 & -11 & 5 & -3 \end{bmatrix} \quad \begin{array}{l} -2R_3 + R_3 \\ -4R_1 + R_4 \end{array}$$

$$\begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 1 & -\frac{5}{11} & \frac{3}{11} \\ 0 & -11 & 5 & -3 \\ 0 & -11 & 5 & -3 \end{bmatrix} \quad \frac{1}{11} R_2$$

$$\begin{bmatrix} 1 & 0 & \frac{4}{11} & \frac{13}{11} \\ 0 & 1 & -\frac{5}{11} & \frac{3}{11} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} -3R_2 + R_1 \\ 11R_2 + R_3 \\ 11R_2 + R_4 \end{array}$$

Required Reduced Echelon Form

$$Q6 \quad A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

$$R \quad \begin{bmatrix} 1 & 2 & 1 \\ 0 & -5 & -1 \\ 0 & 1 & 2 \end{bmatrix} \quad -3R_1 + R_2$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & -5 & -1 \end{bmatrix} \quad R_{23}$$

$$R \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 9 \end{pmatrix}$$

$$5R_2 + R_3 \\ -2R_2 + R_1$$

$$R \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \quad \frac{1}{9} R_3$$

$$R \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$3R_3 + R_1 \\ -2R_3 + R_2$$

Hence $A \sim I_3$

x-----x

Elementary Column Operations.



Equivalent Matrices:

A $m \times n$ matrix B is said to be equivalent to an $m \times n$ matrix A if B can be obtained from A by applying some elementary row & column operations on A . We denote it as $B \sim A$.

If $B \sim A$ then $B = PAQ$.

• where P and Q are non-singular matrices of order m & n .

' P ' is obtained from I_m by some row operations applied on A to get B .

' Q ' is obtained from I_n by the same column operations applied on A to get B .

Normal or Canonical form of a matrix.

A matrix is said to be in normal or canonical form when it has the form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$

where I_r is Identity matrix of order ' r ' & the remaining submatrices are zero matrices. The following are normal matrices.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} I_3 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_1 & 0 \\ 0 & 0 \end{bmatrix}$$

Rank of a Matrix

No of non-zero rows in Echelon or Reduced Echelon Form of a matrix is called Row Rank of a Matrix.

No of non-zero columns in Echelon or reduced Echelon Form of a matrix is called Column Rank of a matrix.

The row rank & the column rank of a matrix are equal.

Q7

$$A = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ 5 & -1 \\ -2 & 3 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ 0 & -16 \\ 0 & 9 \end{bmatrix}$$

$$-5R_1 + R_3$$

$$2R_1 + R_4$$

$$\sim \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 0 & -16 \\ 0 & 9 \end{bmatrix}$$

$$-\frac{1}{2} R_2$$

$$\sim \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$16R_2 + R_3$$

$$-9R_2 + R_4$$

No of non zero Rows = 2

So Rank = 2

$$(ii) A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 1 & 0 \\ -2 & -1 & 3 \\ -1 & 4 & -2 \end{bmatrix}$$

$$R_2 \begin{bmatrix} 1 & 2 & -3 \\ 0 & -3 & 6 \\ 0 & 3 & -3 \\ 0 & 6 & -5 \end{bmatrix} \begin{array}{l} -2R_1 + R_2 \\ +2R_1 + R_3 \\ +R_1 + R_4 \end{array}$$

$$R_2 \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & -2 \\ 0 & 3 & -3 \\ 0 & 6 & -5 \end{bmatrix} -\frac{1}{3} R_2$$

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 3 \\ 0 & 0 & 7 \end{bmatrix} \begin{array}{l} -3R_2 + R_3 \\ -6R_2 + R_4 \end{array}$$

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 7 \end{bmatrix} \frac{1}{3} R_3$$

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} -7R_3 + R_4$$

No. of non zero rows are 3 (1st, 2nd, 3rd)

So Rank = 3

$$(iii) \begin{pmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{pmatrix}$$

$$R \begin{pmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & -3 & -6 & -3 & 3 \\ 0 & -1 & -2 & -1 & 1 \end{pmatrix} \begin{array}{l} \\ -R_1 + R_2 \\ -2R_1 + R_3 \\ -3R_1 + R_4 \end{array}$$

$$R \begin{pmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{array}{l} \\ 3R_2 + R_3 \\ R_2 + R_4 \end{array}$$

Non zero rows are 1st & 2nd

Hence Rank = 2

$$(iv) \begin{pmatrix} 1 & 3 & -2 & 5 & 4 \\ 1 & 4 & 1 & 3 & 5 \\ 1 & 4 & 2 & 4 & 3 \\ 2 & 7 & -3 & 6 & 13 \end{pmatrix}$$

$$R \begin{pmatrix} 1 & 3 & 2 & 5 & 4 \\ 0 & 1 & 3 & -2 & 1 \\ 0 & 1 & 4 & -1 & -1 \\ 0 & 1 & 1 & -4 & 5 \end{pmatrix} \begin{array}{l} \\ -R_1 + R_2 \\ -R_1 + R_3 \\ -2R_1 + R_4 \end{array}$$

$$R \begin{pmatrix} 1 & 3 & -2 & 5 & 4 \\ 0 & 1 & 3 & -2 & 1 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & -2 & -2 & 4 \end{pmatrix}$$

 $-R_2 + R_3$
 $-R_2 + R_4$

$$R \begin{pmatrix} 1 & 3 & -2 & 5 & 4 \\ 0 & 1 & 3 & -2 & 1 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

 $2R_3 + R_4$

3 Non zero Rows 1st, 2nd, 3rd

Hence Rank = 3.

(8)

Matrix A

$$\begin{pmatrix} 1 & -1 & 3 \\ 2 & -4 & 1 \\ 0 & 3 & 2 \end{pmatrix}_{3 \times 3}$$

Row Operation

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Col operation

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \oplus$$

$$R \begin{pmatrix} 1 & -1 & 3 \\ 0 & -2 & -5 \\ 0 & 3 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} -2R_1 + R_2$$

$$R \begin{pmatrix} 1 & -1 & 3 \\ 0 & 1 & -3 \\ 0 & 3 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} R_3 + R_2$$

$$R \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 3 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ -2 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} R_2 + R_1$$

Q.iii) $A = \begin{pmatrix} 1 & -1 & 3 \\ 2 & -4 & 1 \\ 0 & 3 & 2 \end{pmatrix}_{3 \times 3}$

Row Operations

Col operations.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}_{3 \times 3}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{by } -2R_1 + R_2$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & -5 \\ 0 & 3 & 2 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{by } 1C_1 + C_2, -3C_1 + C_3$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 3 & 2 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{by } R_3 + R_2$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 11 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 1 \\ 6 & -3 & -2 \end{pmatrix} \quad \begin{pmatrix} 1 & 1 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad -3R_2 + R_3$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 11 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 1 \\ 6 & -3 & -2 \end{pmatrix} \quad \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \quad 3C_2 + C_3$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 1 \\ 6 & -3 & -2 \end{pmatrix} \quad \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 3/11 \\ 0 & 0 & 1/11 \end{pmatrix} \quad \frac{1}{11} C_3$$

I_3 P Q

Hence Normal form of $\begin{pmatrix} 1 & -1 & 3 \\ 2 & -4 & 1 \\ 0 & 3 & 2 \end{pmatrix}$ is I_3

x ————— x

Normal or Canonical form:-

A matrix is said to be in normal form when it has the form

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

where I_r is identity matrix of order 'r' and remaining submatrices are zero matrices.

e.g. $\left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) = \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{array} \right) = \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$

Q9 Reduce in Canonical form.

i) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

2x2

For Row operation

For col operation

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

by R_{12}

$$\begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

by $-2R_1 + R_2$

$$\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

by $-2C_1 + C_2$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{2}{3} \\ 0 & -\frac{1}{3} \end{bmatrix}$$

by $\frac{1}{3}C_2$

I_2

P

Q

(ii) $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}_{2 \times 3}$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

by R_{12}

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

by $-2R_1 + R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

by $-C_1 + C_2$
by $-C_1 + C_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$-R_2$

Note P, Q are not unique.
i.e. may be diff, if we operate in a different manner.

by $-C_2 + C_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} I_2 & 0 \end{bmatrix} \quad P \quad Q$$

$$\begin{array}{ccc}
 A & I_3 & I_4 \\
 \begin{pmatrix} 1 & 2 & 0 & -1 \\ 3 & 4 & 1 & 2 \\ -2 & 3 & 2 & 5 \end{pmatrix}_{3 \times 4} & \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}_{3 \times 3} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}_{4 \times 4} \\
 \\
 \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & -2 & 1 & 5 \\ 0 & 7 & 2 & 3 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \begin{array}{l} b_1 - 3R_1 + R_2 \\ 2R_1 + R_3 \end{array} \\
 \\
 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 1 & 5 \\ 0 & 7 & 2 & 3 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \begin{array}{l} b_1 - 2C_1 + C_2 \\ 1C_1 + C_4 \end{array} \\
 \\
 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 5 \\ 0 & 2 & 7 & 3 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & -2 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & b_1 C_{23} \\
 \\
 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 11 & -7 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 8 & -2 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & -2 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & b_1 - 2R_2 + R_3 \\
 \\
 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 11 & -7 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 8 & -2 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & -2 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \begin{array}{l} 2C_2 + C_3 \\ -5C_2 + C_4 \end{array} \\
 \\
 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -7 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 8 & -2 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & -2/11 & 1 \\ 0 & 0 & 1/11 & 0 \\ 0 & 1 & 2/11 & -5 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \frac{1}{11} C_3 \\
 \\
 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 8 & -2 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & -14/11 & 7/11 \\ 0 & 0 & 1/11 & -5 \\ 0 & 1 & 2/11 & -5 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \begin{array}{l} b_1 7C_3 + C_4 \\ \frac{11}{11} - 5 = \frac{-4}{11} \end{array} \\
 \\
 \begin{pmatrix} I_3 & 0 \end{pmatrix} & P & Q &
 \end{array}$$

Hence the required normal form is $\begin{pmatrix} I_3 & 0 \end{pmatrix}$

x ————— x

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3.2-2

$$\begin{pmatrix} 1 & -1 & 1 & 1 & 1 \\ 1 & -1 & 2 & 3 & 1 \\ 2 & -2 & 1 & 0 & 2 \\ 1 & 1 & -1 & -3 & 3 \end{pmatrix} \xrightarrow{4 \times 5} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{4 \times 5} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

4x5

$$\begin{pmatrix} 1 & -1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & -1 & -2 & 0 \\ 0 & 2 & -2 & -4 & -2 \end{pmatrix} \xrightarrow{4 \times 5} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ -2 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{4 \times 5} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} R_1+R_2 \\ -2R_1+R_3 \\ -R_1+R_4 \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & -1 & -2 & 0 \\ 0 & 2 & 2 & -4 & -2 \end{pmatrix} \xrightarrow{4 \times 5} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ -2 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{4 \times 5} \begin{pmatrix} 1 & -1 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} C_1+C_2 \\ -C_1+C_3 \\ -C_1+C_4 \\ -C_1+C_5 \end{matrix}$$

PUACP

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & -1 & 0 & -2 & 0 \\ 0 & 2 & 2 & -4 & -2 \end{pmatrix} \xrightarrow{4 \times 5} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ -2 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{4 \times 5} \begin{pmatrix} 1 & -1 & 1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} C_2 \\ C_3 \\ C_4 \\ C_5 \end{matrix}$$

C_{23}

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -8 & -2 \end{pmatrix} \xrightarrow{4 \times 5} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ -3 & 1 & 1 & 0 & 0 \\ 1 & -2 & 0 & 1 & 0 \end{pmatrix} \xrightarrow{4 \times 5} \begin{pmatrix} 1 & -1 & 1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} R_2+R_3 \\ -2R_2+R_4 \end{matrix}$$

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$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -8 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -3 & 1 & 1 & 0 \\ 1 & -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} -2C_2 + C_1 \\ \\ \\ \\ \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & -8 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 0 & 1 \\ -3 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} R_{34} \\ \\ \\ \\ \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -4 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ \frac{1}{2} & -1 & 0 & \frac{1}{2} \\ -3 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} \frac{1}{2}R_3 \\ \\ \\ \\ \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ \frac{1}{2} & -1 & 0 & \frac{1}{2} \\ -3 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 & 1 & 5 & 0 \\ 0 & 0 & 1 & 4 & 1 \\ 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{matrix} 4C_3 + C_4 \\ \\ \\ \\ \end{matrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$